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Antibody Levels and Protection After Hepatitis B Vaccine: Results of a 30-Year Follow-up Study and Response to a Booster Dose

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(See the editorial commentary by Van Damme on pages 1–3.)

Background. The duration of protection in children and adults resulting from hepatitis B vaccination is unknown. In 1981, we immunized a cohort of 1578 Alaska Native adults and children from 15 Alaska communities aged ≥6 months using 3 doses of plasma-derived hepatitis B vaccine.

Methods. Persons were tested for antibody to hepatitis B surface antigen (anti-HBs) levels 30 years after receiving the primary series. Those with levels <10 mIU/mL received 1 booster dose of recombinant hepatitis B vaccine 2–4 weeks later and were then evaluated on the basis of anti-HBs measurements 30 days after the booster.

Results. Among 243 persons (56%) who responded to the original primary series but received no subsequent doses during the 30-year period, 125 (51%) had an anti-HBs level ≥10 mIU/mL. Among participants with anti-HBs levels <10 mIU/mL who were available for follow-up, 75 of 85 (88%) responded to a booster dose with an anti-HBs level ≥10 mIU/mL at 30 days. Initial anti-HBs level after the primary series was correlated with higher anti-HBs levels at 30 years.

Conclusions. Based on anti-HBs level ≥10 mIU/mL at 30 years and an 88% booster dose response, we estimate that ≥90% of participants had evidence of protection 30 years later. Booster doses are not needed.

Keywords. hepatitis B; immunogenicity; Alaska Native; plasma-derived vaccine; Alaska; US; cohort; 30 years; antibody; anti-HBs.

Universal vaccination with hepatitis B vaccine has been very effective at preventing infection with the hepatitis B virus (HBV) and at reducing the development of chronic infection in young children from perinatal or early childhood exposures to HBV [1–6]. However, the duration of protection after vaccination of infants (immunized from birth), older children and adults (with either plasma-derived or recombinant vaccine) is not well understood [7–11]. We previously demonstrated protection out to 22 years among persons vaccinated as children or adults [12–14].

After US licensure of the HBV vaccine in 1981, we immunized a cohort of 1578 Alaska Native adults and children aged ≥6 months with 3 doses of the plasma-derived hepatitis B vaccine [15]. Persons <20 years of age received the 10-μg dose, and adults (aged ≥20 years) received the 20-μg dose. We followed this cohort yearly with HBV serologic testing for the first 11 years and then 15 and 22 years after their first dose of vaccine [12, 14–16]. After 22 years, 60% had a protective level of antibody to hepatitis B surface antigen (anti-HBs; ≥10 mIU/mL), and overall 93% seemed to be protected (had antibody and/or responded to booster dose). In addition, no new acute or chronic HBV infections were found at this time [12].

In this study, conducted 30 years after primary vaccination series, we recruited a subset of the original cohort with the following goals: (1) to determine the proportion of persons with anti-HBs levels ≥10 mIU/mL, (2) to evaluate the immune response to a booster dose of hepatitis B vaccine among those with anti-HBs <10 mIU/mL, and (3) to compare characteristics of persons with or without protective antibody levels.

METHODS

Study Design

This study was performed in a long-term prospective cohort.

Participants

We enrolled participants from 12 of the original 15 communities in the 1981 immunogenicity study with a documented response to the primary plasma-derived hepatitis B vaccine series.

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Persons in the 3 communities who did not participate did not differ from all others in terms of age, sex, and initial antibody level after the primary series. Persons from these 12 communities who had moved to the city of Anchorage or Bethel, the largest town in the region, were also invited to participate. Persons who had received an additional dose of vaccine in the intervening years (except study participants who were received a booster dose at 22 years as part of the study) were excluded.

Communities are remote, and study personnel flew to each of them for 1–2 days of recruiting in the fall of 2011. Participants were notified in advance by letters and phone calls of the date the study team would be present in their community. Persons out of the community or unavailable on the recruitment day were missed. For the 22-year study (in which booster doses were administered to persons with anti-HBs level <10 mIU/mL) [12] we purposely recruited only 7 of the 15 original communities in order to leave 8 communities "naive" for future immunogenicity studies. Among the study participants at 30 years, we had 3 main groups: group 1 comprised persons who did not participate in the 22-year study and therefore had no blood drawn at that time; group 2, persons who did not receive a booster dose at 22 years (because their 22-year anti-HBs level was ≥10 mIU/mL); and group 3, persons (with anti-HBs level <10 mIU/mL) who received a booster dose of vaccine at 22 years.

Laboratory Testing

Serologic specimens from participants were tested at the Arctic Investigations Program Laboratory for antibody to hepatitis B surface antigen (ETI-AB-AUK Plus; DiaSorin) and antibody to hepatitis B core antigen (Ortho HBc ELISA; Ortho Diagnostics), as described elsewhere [12]. A linear standard curve ranging from 5 to 160 mIU/mL was generated with each test run using the immunoglobulin G anti-HBs standard set (ABAU-STD-SET; DiaSorin). The lower limit of detection for the anti-HBs assay was defined as ≥2 mIU/mL. Persons found to be positive for antibody to hepatitis B core antigen were referred to the Alaska Native Medical Center for further testing for hepatitis B surface antigen with enzyme immunoassay and for HBV DNA with the Roche Amplicor Assay.

An anti-HBs level ≥10 mIU/mL was considered protective. Study participants with anti-HBs levels <10 mIU/mL were offered an intramuscular booster dose of 10 μg of hepatitis B vaccine (Recombivax HB; Merck) given to assess immune response to an antigenic challenge. We attempted to determine the concentration of anti-HBs 30 days after the boost (median interval between booster and follow-up blood collection, 33 days). A positive booster dose response was defined as anti-HBs levels ≥10 mIU/mL at 30 days after the boost.

Statistical Analysis

We analyzed the proportion of persons with protective anti-HBs level using the likelihood ratio χ^2 test for categorical variables. Age was analyzed as a categorical variable. The Cochran–Armitage trend test was used to analyze the anti-HBs levels after the HBV primary series. Quantitative anti-HBs levels were also presented as geometric mean concentrations (GMCs). These levels were log-transformed to calculate GMCs, and values <1.0 mIU/mL were assigned a value of 1.0 before transformation. The anti-HBs level 1 month after the HBV booster dose was compared with the level 6 months after the primary HBV vaccine series, using the Wilcoxon signed rank test. Multivariate models were developed for protective anti-HBs level and response to the HBV booster dose using logistic regression. Variables with a univariate P value <.25 were considered in the multivariate model. In addition, after selection of all main effects, all 2-way interactions were evaluated for statistical significance. All statistical analyses were conducted using SAS software (version 9.3; SAS). All P values are 2-sided and are exact when sample sizes necessitated.

Informed Consent

The present study was approved by the Alaska Area Institutional Review Board, the Centers for Disease Control and Prevention (CDC) Institutional Review Board, and 4 Alaska Native tribal health boards, the Yukon-Kuskokwim Health Corporation, the Norton Sound Health Corporation, the Alaska Native Tribal Health Consortium, and the Southcentral Foundation. All participants provided written, informed consent.

RESULTS

Study Cohort

Of 1111 persons available in 15 communities, we chose to recruit from 12 of those communities (8 that did not participate at 22 years and 4 of the largest communities that had participated in the 22-year follow-up). Among the 783 eligible participants, 435 (56%) were recruited (Figure 1). From the 12 communities, there were 342 participants, plus 23 persons who had moved from one of these communities to Bethel and 70 who had moved to Anchorage. The following reasons/categories were given for not participating in the study: the person refused, had moved from the village, did not show up, was lost to follow-up, or was not in community the day of the study.

Study participants and eligible nonparticipants were similar with regard to sex, age, and anti-HBs level after the primary vaccination series (data not shown). Among the 435 study participants, groups 1, 2, and 3, respectively, comprised 243 persons who did not participate in the 22-year study and therefore had no blood drawn at that time, 129 who did not receive a booster vaccine dose at 22 years (owing to anti-HBs levels ≥10 mIU/mL), and 63 who received a booster dose at 22 years (owing to anti-HBs levels <10 mIU/mL). Because group 3 had already received a booster dose and group 2 did not receive a booster dose (anti-HBs ≥10 mIU/mL) at the 22-year follow-up, a detailed description of and analysis of anti-HBs levels at 30 years is restricted to group 1. Response to a booster dose focuses mainly on group 1

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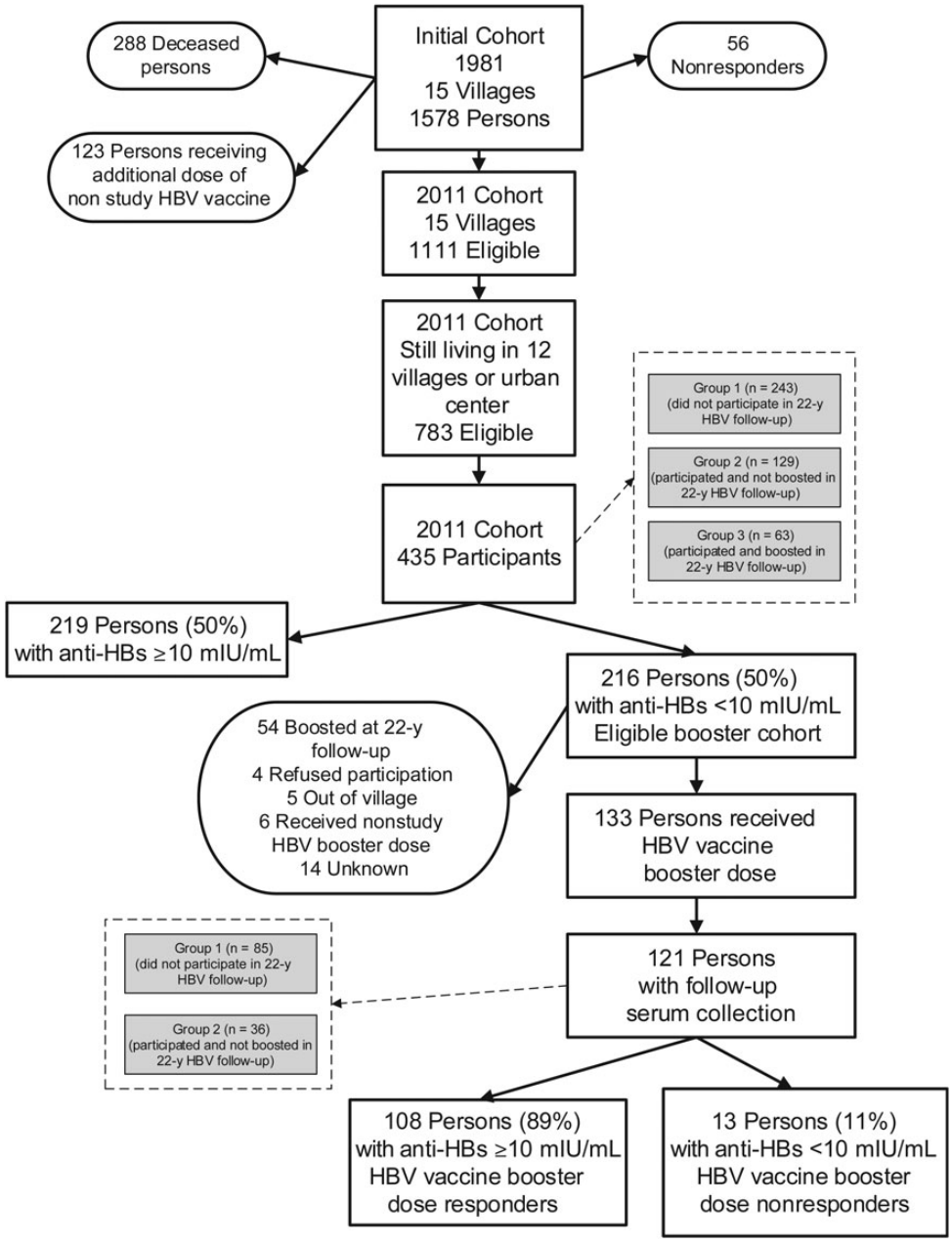


Figure 1. Participant flow chart in a 30-year follow-up study of 1578 persons receiving 3 doses of plasma-derived hepatitis B virus (HBV) vaccine in Alaska in 1981. Abbreviation: anti-HBs, antibody to hepatitis B surface antigen.

but also includes some data on booster dose response for group 2. For group 3, we present only data on immunogenicity 8 years after the booster dose.

Anti-HBs Levels

Overall, among the 435 study participants, 219 (50%) had anti-HBs levels ≥10 mIU/mL; the GMC of anti-HBs was 13.8 mIU/mL. GMCs by group during the 30-year period (Figure 2) highlight the fact that group 3 has had consistently lower GMCs than group 1, and group 2 consistently higher GMCs. Antibody levels at 30 years by sex, age, and anti-HBs level at study enrollment are shown for group 1 (Table 1). Among the 243 group 1 participants, 125 (51%) had anti-HBs levels ≥10 mIU/mL, and the GMC of anti-HBs was 14.4 mIU/mL. Among the 129 group 2 participants, 85 (66%) had anti-HBs levels ≥10 mIU/mL, and the GMC of anti-HBs was 24.6 mIU/mL. Among the 63 study participants in group 3, only 9 (14%) had anti-HBs levels ≥10 mIU/mL 8 years after their booster dose, and the GMC of anti-HBs was 3.6 mIU/mL.

Analysis of data from group 1 by multivariate logistic regression demonstrated that initial anti-HBs level and age at primary

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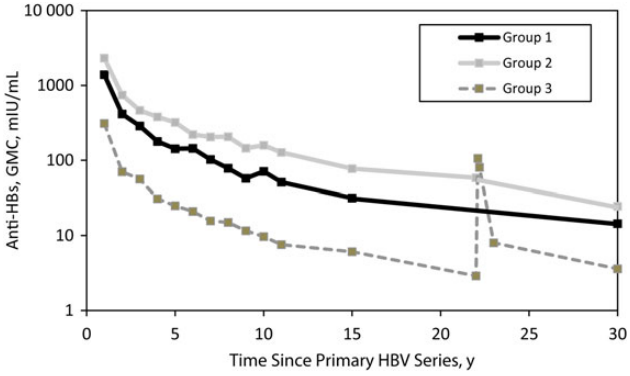


Figure 2. Levels of antibody to hepatitis B surface antigen (anti-HBs) decline over 30 years among 3 groups in Alaska. Group 1 comprised persons who responded to the primary hepatitis B virus (HBV) vaccine series and had not received any subsequent doses; group 2, persons not given a booster dose at a 22-year follow-up (owing to anti-HBs levels ≥10 mIU/mL); and group 3, persons given a booster dose at 22 years (owing to anti-HBs levels <10 mIU/mL). Abbreviation: GMC, geometric mean concentration.

vaccination were associated with anti-HBs levels ≥10 mIU/mL at 30 years. Persons aged 5–19 years at the time of primary vaccination (aged 35–49 years at 30-year follow-up) had higher GMCs and a higher proportion of protective antibodies than study participants in other age groups. No association was found between the proportion of persons with protective antibody levels at 30 years and body mass index, smoking, a history of cancer, or presence of a (self-reported) HBV carrier in the household.

Response to Booster Dose

Because study participants in group 3 received a booster vaccine dose 8 years earlier (22 years after the original vaccine series), they did not receive a booster dose at 30 years. Study participants in groups 1 and 2 with anti-HBs levels <10 mIU/mL received booster doses; however, because group 2 participants had anti-HBs levels ≥10 mIU/mL at 22 years, we chose to focus on group 1 for booster dose response. In group 1, among 118 eligible participants, 96 (81%) who had anti-HBs levels <10 mIU/

Table 1. Level of Anti-HBs 30 Years After Hepatitis B Vaccination, by Sex, Age, and Postvaccination Anti-HBs Level After Primary Series in Persons Who Responded to the Primary Series and Had Not Received Subsequent Doses (Group 1)—Alaska, 2011–2012

Characteristic	Persons, No.[a]	Anti-HBs, GMC, mIU/mL	Anti-HBs ≥10 mIU/mL, % (No.)	*P* Value	Multivariate *P* Value
Sex					
Female	117	14.1	52 (61)	.83	. . .
Male	126	14.7	51 (64)		
Age at 30-y follow-up (age at primary vaccination), y					
<35 (<5)	59	6.8	32 (19)	<.001	.002
35 to <40 (5 to <10)	49	24.9	61 (30)		
40 to <50 (10 to <20)	90	22.8	64 (58)		
≥50 (≥20)	45	8.5	40 (18)		
Anti-HBs level after primary series, mIU/mL					
10 to <100	20	3.4	20 (4)	<.001	<.001
100 to <1000	95	5.8	34 (32)		
1000 to <5000	78	15.0	58 (45)		
≥5000	50	136.9	88 (44)		
BMI, kg/m^2					
<25	68	13.1	53 (36)	.98	. . .
25 to <30	69	17.6	52 (36)		
≥30	101	14.1	51 (52)		
Ever smoked tobacco					
Yes	168	14.1	51 (85)	.48	. . .
No	72	16.6	56 (40)		
Ever chewed tobacco					
Yes	147	14.3	52 (76)	.81	. . .
No	90	16.4	53 (48)		
History of cancer					
Yes	10	39.7	60 (6)	.63	. . .
No	224	14.7	52 (117)		
Self-reported HBV carrier in household					
Yes	7	9.0	57 (4)	>.99	. . .
No	178	15.3	52 (93)		

Abbreviations: Anti-HBs, antibody to hepatitis B surface antigen; BMI, body mass index; GMC, geometric mean concentration; HBV, hepatitis B virus.

[a] No persons had received a transplant, 3 had received immune-modulating medications, 6 had diabetes, 102 currently chewed tobacco, and 93 currently smoked.

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mL received a booster dose of hepatitis B vaccine. The median age of the 85 persons from whom serum was obtained after the booster dose was 40 years; 47 (55%) were male. Among these 85 persons tested 30 days after the booster dose, 75 (88%) responded with anti-HBs levels ≥10 mIU/mL; the GMC was 150.4 mIU/mL (Table 2).

In the multivariate model, we determined that response to a booster dose was associated with the prebooster anti-HBs level; participants with preboost anti-HBs levels of 2–9.9 mIU/mL had a higher probability of achieving a response to booster dose than those with anti-HBs levels <2 mIU/mL ($P = .01$; Table 2). No association was found between the proportion of persons with anti-HBs levels ≥10 mIU/mL after the booster dose and age at primary vaccination, anti-HBs level after primary series, body mass index, smoking, or use of chewing tobacco. In group 2, among the 36 persons tested 30 days after the booster dose, 33 (92%) responded with anti-HBs levels ≥10 mIU/mL; the GMC was 419.8 mIU/mL.

Among the 243 group 1 participants, 125 (51%; 95% confidence interval, 47%–55%) had evidence of long-term protection from hepatitis B vaccination (defined by anti-HBs levels ≥10 mIU/mL). Among the remaining 118 persons with anti-HBs levels <10 mIU/mL, 75 of 85 available participants (88%) who received a booster dose responded with anti-HBs levels ≥10 mIU/mL. Applying the 88% response rate to booster dose to the larger group of 118 persons, 94% of primary responders had evidence of protective antibodies (anti-HBs levels ≥10 mIU/mL) or humoral immune memory defined by response to a booster dose. No new breakthrough infections were seen; however, 2 persons who had previously been identified as positive for antibody to hepatitis B core antigen remained so at 30 years; HBV DNA was not detected.

Table 2. Level of Anti-HBs After Hepatitis B Vaccine Booster Dose in Persons Who Responded to the Primary Series, Had Not Received Subsequent Doses, and Had a 30-Year Anti-HBs Level <10 mIU/mL—Alaska, 2011–2012 (Group 1)

Characteristic	Persons, No.	Anti-HBs, GMC, mIU/mL	Anti-HBs ≥10 mIU/mL, % (No.)	*P* Value	Multivariate *P* Value
Sex					
Female	38	266.2	87 (33)	.75	. . .
Male	47	95.1	89 (42)		
Age at 30-y follow-up (age at primary vaccination), y					
<35 (<5)	30	98.2	83 (25)	.77	. . .
35 to <40 (5 to <10)	17	134.2	94 (16)		
40 to <50 (10 to <20)	19	275.7	89 (17)		
≥50 (≥20)	19	179.4	89 (17)		
Anti-HBs level after primary series, mIU/mL					
10 to <100	9	33.7	89 (8)	.08	. . .
100 to <1000	46	78.9	83 (38)		
1000 to <5000	24	587.7	100 (24)		
≥5000	6	876.3	83 (5)		
BMI, kg/m^2					
<25	23	120.7	87 (20)	.84	. . .
25 to <30	23	185.6	87 (20)		
≥30	36	132.2	89 (32)		
Ever smoked tobacco					
Yes	61	128.4	87 (53)	.72	. . .
No	22	218.9	91 (20)		
Currently smokes tobacco					
Yes	34	129.5	88 (30)	>.99	. . .
No	49	162.1	88 (43)		
Ever chewed tobacco					
Yes	50	167.8	90 (45)	.49	. . .
No	31	117.0	84 (26)		
Currently chews tobacco					
Yes	39	153.4	87 (34)	>.99	. . .
No	46	148.3	89 (41)		
Preboost anti-HBs level, mIU/mL					
<2	39	40.1	82 (32)	.04	.04
2–4	30	222.0	90 (27)		
5–9.9	16	1837.4	100 (16)		

Abbreviations: Anti-HBs, antibody to hepatitis B surface antigen; BMI, body mass index; GMC, geometric mean concentration.

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DISCUSSION

This study, which follows Alaska Native persons who had received plasma-derived hepatitis B vaccine (at >6 months of age) and who responded to the primary vaccine series at that time, is the longest cohort study on long-term protection after hepatitis B vaccination to date in the world. It shows that protection from the vaccine continues out to 30 years; ≥94% of persons had evidence of protection (anti-HBs ≥10 mIU/mL or response to booster dose of vaccine). No chronic infections were documented. In group 1, initial anti-HBs levels after the primary series and age at primary vaccination were correlated with higher anti-HBs levels at 30 years.

Data from this study showing protection from infection with HBV 30 years after vaccine administration are similar to findings from other long-term cohort studies with 20–25 years of follow-up in Taiwan [17], Thailand [18], the Gambia [19], and elsewhere [20, 21]. These studies all included infants, but our original Alaska cohort comprised children and adults and did not include infants, who do not respond to boosting as readily as young children [8, 22]. A higher proportion of persons aged 5–19 years when they received their primary vaccine series had protective antibody levels (anti-HBs ≥10 mIU/mL) 30 years later compared with those aged <5 or >20 years at primary vaccination. Our study findings suggest that children vaccinated through catch-up programs or young adults vaccinated for occupational safety reasons have a high likelihood of being protected for multiple decades.

Among participants in groups 1 and 2 with an anti-HBs level <10 mIU/mL at 30 years, those with a higher preboost anti-HBs level were more likely to demonstrate a booster response (anti-HBs ≥10 mIU/mL) than those with lower preboost anti-HBs levels. This suggests that even if the anti-HBs level at 30 years has waned to <10 mIU/mL, the closer the level is to 10 mIU/mL, the higher the probability that the boost will succeed.

No previous studies, to our knowledge, have defined what constitutes an appropriate booster response in a person who has received the hepatitis B vaccine series in the distant past and whose levels of anti-HBs later fell below the protective level. Although the magnitude of antibody response after a single booster dose is not very vigorous, in our study approximately 90% of persons in groups 1 and 2 who received a booster dose responded with levels of anti-HBs ≥10 mIU/mL. The study done by Szmuness et al [23] in 1980 demonstrated that only 31.4% of persons (naive to this antigen) who were vaccinated with 1 dose of hepatitis B vaccine acquired antibody levels >10 radioimmunoassay units (shown to be equivalent to 10 mIU/mL) [16]. Our data provide strong evidence that the immune system of most persons 30 years after the initial vaccination series continue to recognize this antigen.

The findings from our study also probably pertain to the long-term protection afforded by this vaccine in healthcare workers, who are at a particularly high risk of exposure to and acquisition of HBV if sufficient immunity is not present [24]. The cost of periodic screening of healthcare workers for anti-HBs and boosting those whose levels are low or absent would be prohibitive. Our study provides strong evidence to support the CDC recommendation that protection, after a complete series of hepatitis B vaccine and documentation of the initial response, will last for ≥30 years [24].

Limitations of this study include receipt of plasma-derived vaccine for the participant's primary series because the recombinant vaccine currently in use was not available until the late 1980s. However, studies have shown that antibody response and the effectiveness of the plasma-derived vaccine in Alaska and in other populations is similar to results in studies using the recombinant vaccine [7, 25]. Thus, we believe our findings apply to persons who received either type of vaccine. Another limitation is that the lower limit of detection for the anti-HBs assay is not clear; we chose to define it as ≥2 mIU/mL.

Our study did not include persons vaccinated at birth, in whom anti-HBs levels are known to fall faster than in those immunized starting at >6 months of age [9]. Although this study gives ample evidence that humoral immunity persists for ≥30 years, studies to determine the status of cellular mediated immunity would be complementary and important, especially in those who lose anti-HBs and fail to respond to a booster dose.

In conclusion, we believe that the results of our findings are applicable to children, adolescents, and adults vaccinated during hepatitis B catch-up programs and to healthcare workers vaccinated with either vaccine. Hepatitis booster doses are not currently needed for these groups at 30 years out from primary vaccine series. We plan to follow the antibody and booster response of this unique cohort at 35 years and will also look at cell-mediated immune response among participants.

Notes

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